Week 08 teacher guide

Negative numbers, coordinates, and inequalities

PowerPoint slides 673-764 | Student workbook pp. 62-69

A signed number gives a position relative to zero. Coordinates use two such positions. Inequalities describe all positions that meet a condition.

The seven days

Day 050 · p. 62 · Saturday · 20 questions,including 2 challenges
Negative numbers and distance from zero

Day 051 · pp. 63-64 · Sunday · 20 questions,including 2 challenges
Coordinates and inequalities

Day 052 · p. 65 · Monday · 20 questions,including 2 challenges
Negative numbers, coordinates, and inequalities

Day 053 · p. 66 · Tuesday · 20 questions,including 2 challenges
Negative numbers, coordinates, and inequalities

Day 054 · p. 67 · Wednesday · 25 questions,including 2 challenges
Negative numbers, coordinates, and inequalities

Day 055 · p. 68 · Thursday · 20 questions,including 2 challenges
Negative numbers, coordinates, and inequalities

Day 056 · p. 69 · Friday · 20 questions,including 2 challenges
Weekly check

Use one day at a time

The student pages contain questions only. Work on separate plain paper, labeling the Day number and item IDs. Weekend lessons use 20 core questions including C1 and C2. Weekdays use 20 questions; Wednesday uses 25. Allow 30-45 minutes on ordinary weekdays and 40-55 minutes on Wednesday. Split a demanding day into two sittings if needed.

The Sunday reference page is for checking a method after an attempt. Keep answer-review slides out of view until the corresponding work is complete. Weekly answer-review sections are optional after Friday and do not add to the 80-minute weekend lesson.

Built-in review

Day 052: Number lines and coordinates (9, 10, 11, 12)

Day 053: Number lines and coordinates (9, 10, 11, 12)

Day 054: Weekly puzzle (13)

Day 055: Area, surface area and volume (9, 10, 11, 12)

On Friday, record routine accuracy out of 18 separately from the two challenge explanations. For each challenge, note whether the learner connected the steps, used a valid model or calculation, and justified the conclusion. Revisit a missed model, then check with a fresh item later.

Saturday. Negative numbers and distance from zero

Day 050 · p. 62 | Slides 673-703

Goal: Order positive and negative numbers and describe their distances from zero.

Check first: Fractions and decimals on a number line.

Materials: pencil and plain paper; use the projected models. Sketch strips, grids, or nets as needed; make any manipulatives from plain paper.

80 minutes: recall 7; launch and explore 18; model and guided practice 20; compare methods 10; challenges 20; summary and exit 5. Allow up to 10 extra minutes for discussion. Assessment alternatives replace routine practice.

Follow the numbered slide order below. The timing budgets span the named phases even when guided practice or discussion returns later. On busy weekends, split the lesson into two sittings and resume at the next slide.

Teaching sequence

Slides 673-674 · Orient

Read the goal. Ask what is familiar; name the quantity or relationship in focus.

Slides 675-676 · Recall · R1, R2, R3, R4

Try from memory, then check. Use one error for a quick reminder; keep this brief.

Slides 677-678 · Explore · P1, P2

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 679 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 680-682 · Explore · P3, P4, P5, P6

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 683-685 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 686 · Guided practice · F

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 687 · Compare methods

Connect diagram and calculation. Ask where each quantity appears and why both methods preserve it.

Slides 688 · Discuss an error · D

Locate the first incorrect step. Explain why it is tempting; repair it with a model or counterexample.

Slides 689 · Language

Attach each term to a model. Ask for an example and a non-example.

Slides 690-693 · Guided practice · X1, X2, X3, X4

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 694-695 · Deepening

Use the investigation below. Predict, calculate, and explain the structural reason. Budget about 8 minutes.

Slides 696-699 · Challenges · C1, C2

Budget about 10 minutes per challenge. Require linked reasoning; give a single hint after an independent attempt.

Slides 700 · Summary · S

Ask for one sentence or annotated example in the learner’s own words. Listen for the main relationship.

Slides 701 · Exit · E

Keep answers hidden. Collect an independent response and its reasoning before helping.

Slides 702 · Weekday bridge

Point to the next five days: 18 routines plus 2 challenges, or 23 routines plus 2 challenges on Wednesday.

Slides 703 · Tutor check

After the exit, use the hidden tutor slide to identify secure understanding, a small repair, or needed re-teaching.

Saturday. Discussion and support

Investigation T: Greater number, greater distance?

Watch and discuss; this is a projected investigation, not an extra assigned question.

Two sensors read −3.5°C and −1.25°C. One learner says −3.5 is greater because 3.5 is greater than 1.25. Repair the comparison, compare absolute values, and find the gap between the temperatures. Use one number line to explain all three answers.

1. Place both values left of zero. The rightmost is −1.25.
2. Measure from zero: 3.5 is farther than 1.25.
3. Count the gap from −3.5 to −1.25: 2.25 units.

Ask: Which question asks about position, and which asks about distance?

If stuck: Point rightward to show greater, then measure from zero separately.

Expected: −3.5<−1.25; |−3.5|>|−1.25|; gap 2.25°C.

C1 · Challenge discussion

Ask: How do order and distance from zero answer different questions?

If stuck: Mark zero before comparing the three locations.

Expected: −6<−1<4; −1°C closest to 0; gap 10°C.

C2 · Challenge discussion

Ask: What has to be equal for two locations to be opposites?

If stuck: Split the vertical distance at sea level.

Expected: −12 m, + 5 m; distances 12 m and 17 m; no, the opposite is + 12 m.

What to notice

Locates negatives to the left of zero. → Ask which point is farther right.

Separates a number from its distance. → Ask for both the signed location and the unsigned distance.

Common error: Orders negatives by their digits. Repair: Walk rightward on a number line and read the labels.

Sunday. Coordinates and inequalities

Day 051 · pp. 63-64 | Slides 704-731

Goal: Use coordinates to locate and reflect points, and graph sets of solutions.

Check first: Signed number order and horizontal or vertical distance.

Materials: pencil and plain paper; use the projected models. Sketch strips, grids, or nets as needed; make any manipulatives from plain paper.

80 minutes: recall 7; launch and explore 18; model and guided practice 20; compare methods 10; challenges 20; summary and exit 5. Allow up to 10 extra minutes for discussion. Assessment alternatives replace routine practice.

Follow the numbered slide order below. The timing budgets span the named phases even when guided practice or discussion returns later. On busy weekends, split the lesson into two sittings and resume at the next slide.

Teaching sequence

Slides 704-705 · Orient

Read the goal. Ask what is familiar; name the quantity or relationship in focus.

Slides 706-707 · Recall · R1, R2, R3, R4

Try from memory, then check. Use one error for a quick reminder; keep this brief.

Slides 708-709 · Explore · P1, P2

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 710 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 711-713 · Explore · P3, P4, P5, P6, P7

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 714-716 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 717 · Guided practice · F

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 718 · Compare methods

Connect diagram and calculation. Ask where each quantity appears and why both methods preserve it.

Slides 719 · Discuss an error · D

Locate the first incorrect step. Explain why it is tempting; repair it with a model or counterexample.

Slides 720 · Language

Attach each term to a model. Ask for an example and a non-example.

Slides 721 · Explore · M1, M2, M3

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 722-723 · Deepening

Use the investigation below. Predict, calculate, and explain the structural reason. Budget about 8 minutes.

Slides 724-727 · Challenges · C1, C2

Budget about 10 minutes per challenge. Require linked reasoning; give a single hint after an independent attempt.

Slides 728 · Summary · S

Ask for one sentence or annotated example in the learner’s own words. Listen for the main relationship.

Slides 729 · Exit · E

Keep answers hidden. Collect an independent response and its reasoning before helping.

Slides 730 · Weekday bridge

Point to the next five days: 18 routines plus 2 challenges, or 23 routines plus 2 challenges on Wednesday.

Slides 731 · Tutor check

After the exit, use the hidden tutor slide to identify secure understanding, a small repair, or needed re-teaching.

Sunday. Discussion and support

Investigation T: Two rules, one solution set

Watch and discuss; this is a projected investigation, not an extra assigned question.

A temperature t is acceptable if it is at least −2°C and less than 3°C. Compare the two inequalities $-2 \le t < 3$ and $-2 < t \le 3$. List the integer solutions to each, identify exactly where they differ, and draw the endpoint choices.

1. Translate at least as greater than or equal to, and less than as strictly less.
2. Test −2 and 3 directly; then list the interior integers.
3. Compare the two endpoint patterns: interior points are unchanged.

Ask: Which two values do the different endpoint symbols affect?

If stuck: Test each boundary directly in both rules before marking circles.

Expected: First: −2, −1, 0, 1, 2. Second: −1, 0, 1, 2, 3. They differ only at −2 and 3.

C1 · Challenge discussion

Ask: Which coordinates change, and which distances remain the same?

If stuck: Count each horizontal and vertical leg separately.

Expected: 12 units; A′(4, 2), B′(−3, 2), C′(−3, −3); 12 units.

C2 · Challenge discussion

Ask: Which words decide whether an endpoint is included?

If stuck: List the first rule's ages, then test the second rule.

Expected: $10 \le a < 14$ and $a \le 12$; both allow 10, 11, 12. 10 and 12 included; 14 excluded.

What to notice

Moves horizontally before vertically. → Have the student trace each move with a finger.

Preserves distance from the reflection axis. → Count equal squares on each side of the axis.

Common error: Swaps the coordinates. Repair: Use horizontal then vertical every time.

Slide-order index

Follow this numerical order. The session plans explain how to teach each phase; full scripts remain in PowerPoint speaker notes. Item IDs match the student workbook.

Saturday

673. Week 8, Saturday
Watch and discuss

674. Today
Watch and discuss

675. Recall
Day 050 · Workbook p. 62 · R1, R2, R3, R4

676. Recall: answers
Day 050 · Workbook p. 62 · R1, R2, R3, R4

677. Temperature
Day 050 · Workbook p. 62 · P1

678. Integers on a number line
Day 050 · Workbook p. 62 · P2

679. The model
Watch and discuss

680. Fraction and decimal negative numbers
Day 050 · Workbook p. 62 · P3

681. Opposite
Day 050 · Workbook p. 62 · P4, P5

682. Number order versus absolute-value order
Day 050 · Workbook p. 62 · P6

683. Worked example · 1 of 3
Watch and discuss · worked example

684. Worked example · 2 of 3
Watch and discuss · worked example

685. Worked example · 3 of 3
Watch and discuss · worked example

686. Your turn
Day 050 · Workbook p. 62 · F

687. Two ways to see it
Watch and discuss · compare the same example

688. A claim to check
Day 050 · Workbook p. 62 · D

689. Words and meaning
Watch and discuss

690. Guided practice · X1
Day 050 · Workbook p. 62 · X1

691. Guided practice · X2
Day 050 · Workbook p. 62 · X2

692. Guided practice · X3
Day 050 · Workbook p. 62 · X3

693. Guided practice · X4
Day 050 · Workbook p. 62 · X4

694. Greater number, greater distance?
Watch and discuss · draw on separate paper

695. Greater number, greater distance?
Watch and discuss · draw on separate paper

696. Challenge C1
Day 050 · Workbook p. 62 · C1

697. Challenge C1 · Discuss
Day 050 · Workbook p. 62 · C1

698. Challenge C2
Day 050 · Workbook p. 62 · C2

699. Challenge C2 · Discuss
Day 050 · Workbook p. 62 · C2

700. In your own words
Day 050 · Workbook p. 62 · S

701. Exit ticket
Day 050 · Workbook p. 62 · E

702. The week ahead
Days 052-056 · Workbook pp. 65-69

703. Tutor reference
Tutor only;hidden in slide show

Sunday

704. Week 8, Sunday
Watch and discuss

705. Today
Watch and discuss

706. Recall
Day 051 · Workbook p. 63 · R1, R2, R3, R4

707. Recall: answers
Day 051 · Workbook p. 63 · R1, R2, R3, R4

708. Plot points in all four quadrants
Day 051 · Workbook p. 63 · P1

709. Read coordinates from a graph
Day 051 · Workbook p. 63 · P2

710. The model
Watch and discuss

711. Reflect across each axis
Day 051 · Workbook p. 63 · P3, P4

712. Polygon perimeter on coordinate plane
Day 051 · Workbook p. 63 · P5

713. Open and closed circles
Day 051 · Workbook p. 63 · P6, P7

714. Worked example · 1 of 3
Watch and discuss · worked example

715. Worked example · 2 of 3
Watch and discuss · worked example

716. Worked example · 3 of 3
Watch and discuss · worked example

717. Your turn
Day 051 · Workbook p. 63 · F

718. Two ways to see it
Watch and discuss · compare the same example

719. A claim to check
Day 051 · Workbook p. 63 · D

720. Words and meaning
Watch and discuss

721. Mixed review
Day 051 · Workbook p. 63 · M1, M2, M3

722. Two rules, one solution set
Watch and discuss · draw on separate paper

723. Two rules, one solution set
Watch and discuss · draw on separate paper

724. Challenge C1
Day 051 · Workbook p. 63 · C1

725. Challenge C1 · Discuss
Day 051 · Workbook p. 63 · C1

726. Challenge C2
Day 051 · Workbook p. 63 · C2

727. Challenge C2 · Discuss
Day 051 · Workbook p. 63 · C2

728. In your own words
Day 051 · Workbook p. 63 · S

729. Exit ticket
Day 051 · Workbook p. 63 · E

730. The week ahead
Days 052-056 · Workbook pp. 65-69

731. Tutor reference
Tutor only;hidden in slide show

Answer review

732. Week 8 answers
Next lesson: slide 765. Use this review section after Friday.

733. Negative numbers and distance from zero · 1 of 5
Day 050 · Workbook p. 62 · Items R1, R2, R3, R4

734. Negative numbers and distance from zero · 2 of 5
Day 050 · Workbook p. 62 · Items P1, P2, P3, P4

735. Negative numbers and distance from zero · 3 of 5
Day 050 · Workbook p. 62 · Items P5, P6, F, D, X1

736. Negative numbers and distance from zero · 4 of 5
Day 050 · Workbook p. 62 · Items X2, X3, X4, S

737. Negative numbers and distance from zero · 5 of 5
Day 050 · Workbook p. 62 · Items E, C1, C2

738. Coordinates and inequalities · 1 of 5
Day 051 · Workbook p. 63 · Items R1, R2, R3, R4, P1

739. Coordinates and inequalities · 2 of 5
Day 051 · Workbook p. 63 · Items P2, P3, P4, P5

740. Coordinates and inequalities · 3 of 5
Day 051 · Workbook p. 63 · Items P6, P7, F, D

741. Coordinates and inequalities · 4 of 5
Day 051 · Workbook p. 63 · Items M1, M2, M3, S

742. Coordinates and inequalities · 5 of 5
Day 051 · Workbook p. 63 · Items E, C1, C2

743. Negative numbers, coordinates, and inequalities · 1 of 4
Day 052 · Workbook p. 65 · Items 1, 2, 3, 4, 5

744. Negative numbers, coordinates, and inequalities · 2 of 4
Day 052 · Workbook p. 65 · Items 6, 7, 8, 9, 10

745. Negative numbers, coordinates, and inequalities · 3 of 4
Day 052 · Workbook p. 65 · Items 11, 12, 13, 14, 15

746. Negative numbers, coordinates, and inequalities · 4 of 4
Day 052 · Workbook p. 65 · Items 16, 17, 18, C1, C2

747. Negative numbers, coordinates, and inequalities · 1 of 5
Day 053 · Workbook p. 66 · Items 1, 2, 3, 4, 5

748. Negative numbers, coordinates, and inequalities · 2 of 5
Day 053 · Workbook p. 66 · Items 6, 7, 8, 9, 10

749. Negative numbers, coordinates, and inequalities · 3 of 5
Day 053 · Workbook p. 66 · Items 11, 12, 13, 14, 15

750. Negative numbers, coordinates, and inequalities · 4 of 5
Day 053 · Workbook p. 66 · Items 16, 17, 18, C1

751. Negative numbers, coordinates, and inequalities · 5 of 5
Day 053 · Workbook p. 66 · Items C2

752. Negative numbers, coordinates, and inequalities · 1 of 5
Day 054 · Workbook p. 67 · Items 1, 2, 3, 4, 5

753. Negative numbers, coordinates, and inequalities · 2 of 5
Day 054 · Workbook p. 67 · Items 6, 7, 8, 9, 10

754. Negative numbers, coordinates, and inequalities · 3 of 5
Day 054 · Workbook p. 67 · Items 11, 12, 13, 14, 15

755. Negative numbers, coordinates, and inequalities · 4 of 5
Day 054 · Workbook p. 67 · Items 16, 17, 18, 19, 20

756. Negative numbers, coordinates, and inequalities · 5 of 5
Day 054 · Workbook p. 67 · Items 21, 22, 23, C1, C2

757. Negative numbers, coordinates, and inequalities · 1 of 4
Day 055 · Workbook p. 68 · Items 1, 2, 3, 4, 5

758. Negative numbers, coordinates, and inequalities · 2 of 4
Day 055 · Workbook p. 68 · Items 6, 7, 8, 9, 10

759. Negative numbers, coordinates, and inequalities · 3 of 4
Day 055 · Workbook p. 68 · Items 11, 12, 13, 14, 15

760. Negative numbers, coordinates, and inequalities · 4 of 4
Day 055 · Workbook p. 68 · Items 16, 17, 18, C1, C2

761. Weekly check · 1 of 4
Day 056 · Workbook p. 69 · Items 1, 2, 3, 4, 5

762. Weekly check · 2 of 4
Day 056 · Workbook p. 69 · Items 6, 7, 8, 9, 10

763. Weekly check · 3 of 4
Day 056 · Workbook p. 69 · Items 11, 12, 13, 14, 15

764. Weekly check · 4 of 4
Day 056 · Workbook p. 69 · Items 16, 17, 18, C1, C2

Complete weekly answers

Match Day,item ID,and page before checking. Equivalent correct forms and valid alternative methods are welcome. Computational keys are compact; challenges and conceptual items include reasoning. No challenge is complete with an unsupported answer alone.

Day 050 · Negative numbers and distance from zero

Workbook p. 62 · assigned set

R1 · D-ADD

8.528

R2 · E-ADD

x = 7

R3 · G-PARA

48 square cm

R4 · D-MULT

$7.35

P1 · N-CONTEXT

−4°C; + 7 m; −$9. Zero means 0°C, sea level, or no money owed or held.

P2 · N-ORDER

Left to right: −6, −2, 1; least −6

Reason: The leftmost point is the least.

P3 · N-ORDER

$-2.5, -\frac{3}{4}, 0.5$

P4 · N-OPP

9; −9; −(−9) = 9

P5 · N-ABS

8, 8, 0 units from zero

Reason: Both −8 and 8 lie eight units from zero.

P6 · N-ABS

−9 < −4; 9 > 4; −4°C is warmer

Reason: −4 is farther right, but −9 is farther from zero.

F · N-ABS

7; 7; 7

Reason: Check each step against the projected worked example; require the learner to explain the operation or relationship.

D · N-ORDER

−10 < −3 because −10 lies farther left.

Reason: Uses whole-number order without considering direction. Find zero, then trace left to each point.

X1 · N-CONTEXT

−3, + 2; sea level.

X2 · N-ORDER

-3, -1, 0, $\frac{3}{2}$

X3 · N-OPP

3, then −3.

Reason: Each reflection across zero reverses the sign; two reflections return to the starting point.

X4 · N-ABS

3, 3, 0; equal.

Reason: Absolute value measures distance from zero, so opposite numbers have equal absolute values.

S · EXPLAIN

Answers vary. Numbers increase to the right. Absolute value measures distance from zero, regardless of direction.

Reason: Accept an accurate explanation supported by a matching example.

E · N-ORDER

$-6, -2.5, -\frac{1}{4}, 1.5$; number −6, absolute value 6

Reason: Place all four points, then compare their distances from zero.

C1 · N-ABS

−6<−1<4; −1°C closest to 0; gap 10°C.

Reason: Position the values on one line. The distance from −6 to 0 is 6 and from 0 to 4 is 4, totaling 10. The largest absolute value is 6, but −6 is the coldest value because it lies farthest left.

C2 · N-OPP

−12 m, + 5 m; distances 12 m and 17 m; no, the opposite is + 12 m.

Reason: Zero marks sea level. The full vertical trip is 12 m up to zero and 5 m farther, totaling 17. Opposite positions must have equal distances from zero, so + 5 is not opposite −12.

Day 051 · Coordinates and inequalities

Workbook p. 63 · assigned set

R1 · D-DIV

20

R2 · E-MULT

x = 25

R3 · G-SA

62 square cm

R4 · D-SUB

7.714

P1 · N-COORD

A I; B II; C III; D IV

Reason: Read x first, then y. Their signs identify the quadrant.

P2 · N-COORD

P(−2, 4); Q(3, −1)

P3 · N-REFLECT

Across x-axis: (−2, −3). Across y-axis: (2, 3).

P4 · N-DIST

7 units; 7 units

P5 · N-DIST

16 units

P6 · E-INEQ

x > 2: open circle, shade right, excludes 2. x ≥ 2: closed circle, shade right, includes 2.

Reason: The equality bar decides whether the endpoint belongs.

P7 · E-INEQ

h ≥ 48; yes

F · N-REFLECT

(5, −2); 10 units

Reason: Check each step against the projected worked example; require the learner to explain the operation or relationship.

D · E-INEQ

Use x ≥ 6, a closed circle at 6, and shade right.

Reason: Treats at least as more than. Test the exact boundary value in the words.

M1 · D-ADD

4.528

M2 · E-EVAL

13

M3 · G-AREA

16 square cm

S · EXPLAIN

Answers vary. A coordinate pair gives horizontal then vertical position. An inequality can describe many solutions.

Reason: Accept an accurate explanation supported by a matching example.

E · N-REFLECT

A′(4, 1.5); distance 8 units

Reason: The y-coordinate stays 1.5. Both points are 4 units from the y-axis, so the gap is 8.

C1 · N-REFLECT

12 units; A′(4, 2), B′(−3, 2), C′(−3, −3); 12 units.

Reason: The horizontal leg is 7 and vertical leg 5, totaling 12. Reflecting changes every x-sign, while y stays fixed. The new horizontal gap is still 7 and vertical gap 5, so total length is unchanged.

C2 · E-INEQ

10≤a<14 and a≤12; both allow 10, 11, 12. 10 and 12 included; 14 excluded.

Reason: The first rule allows 10, 11, 12, 13. The second keeps only values no greater than 12. Their overlap is 10, 11, 12. At least and at most include the stated boundary; less than excludes 14.

Day 052 · Negative numbers, coordinates, and inequalities

Workbook p. 65 · assigned set

1 · N-FACT

30

2 · N-FACT

70

3 · N-FACT

40

4 · N-FACT

90

5 · G-PARA

48 square cm

6 · P-PERCENT

75 %

7 · N-FACT

24

8 · N-ABS

5 units from zero

Reason: Count the unit steps between the number and zero.

9 · N-ORDER · review
−9, −4, 0

10 · N-ABS · review
12 units from zero

Reason: Count the unit steps between the number and zero.

11 · N-ABS · review
0 units from zero

Reason: Count the unit steps between the number and zero.

12 · N-OPP · review
6

13 · N-ORDER
−10 < −3 because −10 lies farther left.

14 · N-CONTEXT
−5, + 4; sea level.

15 · N-ORDER
-5, -2, 0, $\frac{3}{2}$

16 · N-OPP
5, then −5.

Reason: Each reflection across zero reverses the sign; two reflections return to the starting point.

17 · N-ABS
5, 5, 0; equal.

Reason: Absolute value measures distance from zero, so opposite numbers have equal absolute values.

18 · N-ABS
−5°C warmer; −7 has greater absolute value.

Reason: −5 is farther right, but −7 is farther from zero.

C1 · N-DIST
Opposite + 7; whole 14 km; parts 5 km and 9 km.

Reason: The opposite of −7 is + 7. Crossing zero gives 7 + 7 = 14. From −7 to −2 is 5; from −2 to + 7 is 2 + 7 = 9. The parts 5 + 9 recover 14.

C2 · N-ABS
−7; opposite 7; −7<7 and |−7| = |7| = 7.

Reason: Ten units left of 3 reaches −7; ten units right reaches 13 but is not negative. The opposite is 7. Both are 7 units from zero, although 7 is greater in number-line order.

Day 053 · Negative numbers, coordinates, and inequalities

Workbook p. 66 · assigned set

1 · F-EQUIV
0.5; 50%

2 · F-EQUIV
0.25; 25%

3 · F-EQUIV
0.75; 75%

4 · F-EQUIV
0.125; 12.5%

5 · G-SA
62 square cm

6 · F-QUOT
$\frac{5}{6}$ of a bar

7 · N-FACT
12

8 · N-DIST
9 units

9 · N-REFLECT · review
(3, 4)

10 · N-DIST · review
7 units

11 · E-INEQ · review
x ≥ 9; yes

12 · E-INEQ · review
x < 5; no

13 · E-INEQ
Use x ≥ 6, a closed circle at 6, and shade right.

14 · N-CONTEXT
−6, + 5; sea level.

15 · N-ORDER
-6, $\frac{-5}{2}$, 0, $\frac{3}{2}$

16 · N-OPP
6, then −6.

Reason: Each reflection across zero reverses the sign; two reflections return to the starting point.

17 · N-ABS

6, 6, 0; equal.

Reason: Absolute value measures distance from zero, so opposite numbers have equal absolute values.

18 · N-ABS

−6°C warmer; −8 has greater absolute value.

Reason: −6 is farther right, but −8 is farther from zero.

C1 · N-COORD

Sides 6 and 5; perimeter 22; area 30; (−5, −3); none of these measurements change.

Reason: Horizontal distance 1−(−5) = 6; vertical distance 3−(−2) = 5. Perimeter 2(6 + 5) = 22 and area 6 × 5 = 30. Reflection reverses y-signs but preserves lengths and area.

C2 · E-INEQ

1.5 + 2w≤8; 3 and 3.25 fit; 3.5 does not.

Reason: Include the box and both items. The totals are 7.5, 8, and 8.5 kg. At most permits exactly 8. The largest tested fitting item is 3.25 kg.

Day 054 · Negative numbers, coordinates, and inequalities

Workbook p. 67 · assigned set

1 · N-SCALE

102

2 · N-SCALE

10

3 · N-SCALE

21

4 · N-SCALE

30

5 · N-ABS

0 units from zero

Reason: Count the unit steps between the number and zero.

6 · N-OPP

–9

7 · G-PARA

24 square cm

8 · G-COMP

37 square cm

9 · F-SCALE

Larger; 8

Reason: Four thirds takes one whole six plus one third of six.

10 · P-PART

6 stickers

11 · N-GCF

Common factors: 1, 2, 3, 6; GCF 6

12 · N-GCF

Common factors: 1, 2, 4; GCF 4

13 · N-REFLECT · review

Start (−3, 2); after y-axis (3, 2); after x-axis (3, −2)

14 · N-CONTEXT

−7, + 2; sea level.

15 · N-ORDER

-7, -1, 0, $\frac{3}{2}$

16 · N-OPP

7, then −7.

Reason: Each reflection across zero reverses the sign; two reflections return to the starting point.

17 · N-ABS

7, 7, 0; equal.

Reason: Absolute value measures distance from zero, so opposite numbers have equal absolute values.

18 · N-ABS

−7°C warmer; −9 has greater absolute value.

Reason: −7 is farther right, but −9 is farther from zero.

19 · N-DIST

9

20 · N-COORD

A: I; B: II; C: III; D: IV.

21 · N-REFLECT

Across x: (−2, −7); across y: (2, 7).

22 · N-DIST

9 units.

Reason: Equal y-coordinates put the segment on a horizontal line; count the x-gap.

23 · N-DIST

24 units.

C1 · N-DIST

B = (−11, −2) or (5, −2); quadrant IV point (5, −2); reflection (−5, −2).

Reason: Move 8 units left or right from x = −3 while y remains −2. This gives x = −11 or 5. Quadrant IV has positive x and negative y; reflecting across the y-axis reverses only x.

C2 · N-GCFLCM

36 seconds; 12-second light 3 flashes, 18-second light 2 flashes.

Reason: The first positive shared multiple is 36. The first light flashes at 12, 24, 36; the second at 18, 36. GCF 6 is a shared divisor, not a time occurring on both flash lists.

Day 055 · Negative numbers, coordinates, and inequalities

Workbook p. 68 · assigned set

1 · N-EST

1000

2 · N-EST

11

3 · N-EST

About 100

4 · N-EST

About 20

5 · N-ABS

0 units from zero

Reason: Count the unit steps between the number and zero.

6 · N-OPP

–9

7 · N-REFLECT

(3, 4)

8 · G-TRI

18 square cm

9 · G-AREA · review

21 square cm

10 · G-COMP · review

38 square cm

11 · G-SA · review

66 square cm

12 · G-VOL · review

$\frac{2}{3}$ cubic meters

13 · E-INEQ

$x \geq 9$; yes

14 · E-INEQ

Use $x \geq 6$, a closed circle at 6, and shade right.

15 · N-SIGNED

2

16 · E-EQ

$x = 4$

17 · N-CONTEXT

−8, + 3; sea level.

18 · N-ORDER

$-8, \frac{-3}{2}, 0, \frac{3}{2}$

C1 · E-INEQ

Allowed −8 and −5; $-8 \leq t < -3$.

Reason: At least −8 includes the left boundary. Below −3 excludes the right boundary. The allowed points lie to the right of or at −8 and strictly left of −3.

C2 · N-COORD

(−2, −3), then (4, −3); 13 units; yes, both finish (4, −3).

Reason: Vertical movement changes only y: 4 to −3. Horizontal movement changes only x: −2 to 4. Reversing the order reaches (4, 4) first and then (4, −3). Both routes have lengths 7 + 6 = 13.

Day 056 · Weekly check

Workbook p. 69 · assigned set

1 · N-ABS

0 units from zero

Reason: Count the unit steps between the number and zero.

2 · N-OPP

–9

3 · N-REFLECT

(3, 4)

4 · N-ABS

12 units from zero

Reason: Count the unit steps between the number and zero.

5 · E-MULT

$x = 9$

6 · E-MULT

$x = 3.2$

7 · G-SA

40 square cm

8 · F-SCALE

Larger; 8

Reason: Four thirds takes one whole six plus one third of six.

9 · P-PART

6 stickers

10 · N-ORDER

$-10 < -3$ because -10 lies farther left.

11 · N-CONTEXT

-9, $+4$; sea level.

12 · N-ORDER

-9, -2, 0, $\frac{3}{2}$

13 · N-OPP

9, then -9.

Reason: Each reflection across zero reverses the sign; two reflections return to the starting point.

14 · N-ABS

9, 9, 0; equal.

Reason: Absolute value measures distance from zero, so opposite numbers have equal absolute values.

15 · N-ABS

-9°C warmer; -11 has greater absolute value.

Reason: -9 is farther right, but -11 is farther from zero.

16 · N-DIST

13

17 · N-COORD

A: I; B: II; C: III; D: IV.

18 · N-REFLECT

Across x: $(-4, -9)$; across y: $(4, 9)$.

C1 · N-ABS

Bonus -4, -3, 3; opposites 4, 3, -3; only 3 and -3 remain allowed.

Reason: Test the original integer interval. Values at least 3 units from zero are -4, -3, 3. Reversing signs gives 4, 3, -3, but 4 exceeds the original upper bound 3.

C2 · N-REFLECT

Perimeter 20 units; image leftmost $x = -3$.

Reason: The side length is $3-(-2) = 5$, so perimeter 20. Reflection changes x-values -2 and 3 into 2 and -3. Every horizontal position changes sign, so a negative x becomes positive.